Lesson 7-1 · Session 1

Integers and Opposites

Name _____ Date _____ Period _____

Maryland Standard 6.NOS.5 · Learning target — I can write an integer for a real situation and name its opposite.

Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- Integers are positive and negative whole numbers, and zero.
- Opposite integers are the same distance from 0 but on different sides (e.g., 5 and −5).
- Watch out — a negative sign shows direction from zero, not that a quantity is wrong.
- Check it — a number and its opposite are always the same distance from 0.

WORKED EXAMPLE

Writing an Integer and Naming Its Opposite A submarine sits 47 meters below the surface of the water. Write an integer for its depth, then name the opposite.

Step 1 Decide the direction. Below the surface is less than 0, so the integer is negative.

Step 2 Write the integer with a negative sign. depth = −47

Step 3 The opposite sits the same distance from 0 on the other side. 47

Step 4 Check on a number line: one point is left of 0, one is right, both 47 units away.

Answer The depth is −47 meters and its opposite is 47.

Practice

1. Write the opposite of 8. _____ Write the opposite of −12. _____

Answer _____

SHOW YOUR THINKING

2. A submarine is at −150 feet (150 feet below sea level). Write this situation using an integer, then describe its opposite.

Answer _____

RECORD YOUR RESPONSE

3. Plot 4 and its opposite on a number line.

Number line. Use the line below.



Answer _____

SHOW YOUR THINKING

4. Write each answer on the line.

a. The opposite of 21 is _____
b. The opposite of −16 is _____
c. The opposite of 0 is _____
d. Write an integer for 34 feet below sea level. _____
e. Write an integer for a gain of 9 yards. _____

Answer _____

SHOW YOUR THINKING

5. Roof Deck. A repair log lists a roof deck 22 feet above the ground floor. Write an integer for the roof deck, then write the integer for a room at the opposite height.

Answer _____

SHOW YOUR THINKING

6. Locker Fund. The sixth grade fund gained 45 dollars at a bake sale. The next week it lost 45 dollars on supplies. Write an integer for each week, then tell what a total of 0 would mean for the fund.

Answer _____

SHOW YOUR THINKING

7. Plot −6 and its opposite on a number line from −8 to 8. Then write one sentence telling how far each point sits from 0.

Number line. Use the line below.



Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

8. Tariq says the opposite of −30 must still be negative, because the number started out negative. Explain the mistake, then write the correct opposite.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. A diver is 32 feet below sea level. Write the integer and its opposite.

Answer _____ integer and opposite

SHOW YOUR THINKING

10. Write the integer for a withdrawal of $60 and the integer for a deposit of $60.

Answer _____ both integers

SHOW YOUR THINKING

11. Name the Opposites. Give the opposite of −41, of 27, and of 0.

Answer _____ all three

RECORD YOUR RESPONSE

12. Find the Mistake. Jules says the opposite of −8 is −8 because the sign stays. Explain the mistake.

Answer _____ the mistake

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 6-8 Session 1. Simplify 2(n + 5) + 4n.

Answer _____

SHOW YOUR THINKING

14. From Lesson 6-3 Session 2. Find the value of 3 to the 2nd power.

Answer _____

SHOW YOUR THINKING

15. From Lesson 4-4 Session 1. Find 45% of 300.

Answer _____

SHOW YOUR THINKING

Test practice

16. A loss of 14 yards is written as —

□ A. 41	□ B. 14
□ C. 0	□ D. −14

17. The opposite of 23 is —

□ A. −23	□ B. 23
□ C. 0	□ D. 32

18. Numbers that are opposites are the same distance from —

- □ A. one
- □ B. zero
- □ C. each other's sign
- □ D. the largest value

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Zero is its own opposite. A negative number has a positive opposite. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Check signs, scale, and labels. Use a number line or coordinate grid to verify one response.